IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

OPTICAL COMMUNICATION

Thursday 18 May 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question ONE, and ANY THREE of Questions 2 to 5

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S. Lucyszyn

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Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge : $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for TE modes in a symmetric slab waveguide of thickness d are

$\kappa = k_{1x} \tan(k_{1x} d/2)$ and $\kappa = -k_{1x} \cot(k_{1x} d/2)$

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value.

[20]

a) Calculate the time needed for an optical pulse to travel through 250 km of a silica-based optical fibre.

b) A slab waveguide supports 3 transverse electric (TE) modes, $m = 0$, 1 and 2, for a nominal wavelength λ_o of 1.10 μm. At least how many modes could be supported for $\lambda_o = 0.55$ μm?

c) An optical fibre with a silica core has a numerical aperture of 0.20. Calculate the refractive index difference Δn.

d) Which type of noise in optical communications cannot be reduced by employing an optical amplifier? Provide a brief justification.

e) A photodiode produces 16 nA of photocurrent for an incident optical power of 33 nW at a nominal wavelength $\lambda_o = 663$ nm. Calculate the quantum efficiency η of the diode.

f) A certain symmetric slab waveguide of thickness $d = 6$ μm supports exactly 3 TE modes for $\lambda = 750$ nm. Calculate the range of possible values of numerical aperture (NA) of this waveguide.

g) The wavelength of zero dispersion is different in silica fibre and in bulk silica glass. Provide a brief explanation.

h) In semiconductors, for small electric field E values, the drift velocity is proportional to E and another quantity. Provide the name and the units of that quantity. Is this quantity higher for electrons or for holes?

i) A steady electromagnetic wave of 10^{11} photons per second at $\lambda_o = 760$ nm arrives at an optical receiver. Calculate the power level of the optical signal in both nW and dBm.

j) The optical absorption of a particular glass material results in 8% intensity loss at a distance of 250 m. What is the attenuation coefficient of this material in dB/km.

2. a) The symmetric slab waveguide shown in Fig. 2.1 is operating at a given nominal wavelength λ_o. For the lowest order transverse electric (TE) mode (m=0), the electric field amplitude at the core boundary is found to be 0.5 times that at the centre, i.e. $E(x=d/2) = 0.5\ E(x=0)$. Calculate the value of $X = k_{1x}d/2$ for this mode.

[4]

b) Calculate the number of TE modes supported by the waveguide of part (a). Plot a diagram of Y vs X (with $Y = \kappa d/2$), showing the supported modes.

[4]

c) Show that for the m=0 mode described in (a), the effective index n' satisfies the relation: $n'^2 = (3n_1^2 + n_2^2)/4$

[6]

d) If the m=0 mode has unit electric field amplitude (in V/m) at x=0, find the amplitude at $x = 2d$.

[6]

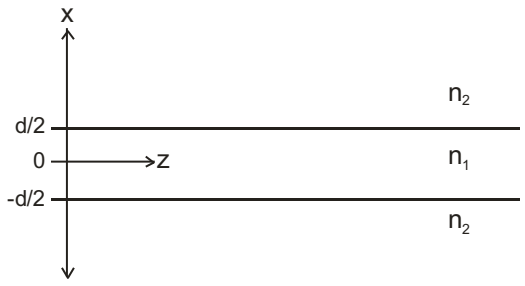


Figure 2.1 Slab waveguide

3. a) A certain Fabry-Perot laser diode has a cavity length of 750 μm and an effective index for the cavity modes of 3.35. The nominal wavelength is 1150 nm. Find the spacing between the nominal wavelengths of the longitudinal modes in nm.

[4]

b) For the laser in (a), estimate the number of longitudinal modes in the spectrum.

[4]

c) The laser is then modified by adding distributed Bragg reflectors. Calculate a suitable period Λ for these. What is the main advantage of such structures?

[4]

d) Using a quantum efficiency of 0.75, calculate the slope efficiency S of this laser. Hence, if a small sinusoidal current of amplitude 10 μA is added to a suitable bias current to bring the laser above threshold, what will be the amplitude of the sinusoidal component of the output optical power?

[4]

e) Explain why the maximum modulation rate of laser diodes can be much higher than that of an LED.

[4]

4. a) Two optical links are constructed over the same length. Link A uses a receiver with noise equivalent power (NEP) of 12.0 pW/√Hz, and fibre with loss 0.45 dB/km. Link B uses a receiver with NEP of 9.0 pW/√Hz, and fibre with loss 0.47 dB/km. The transmitted power and bit rate are the same for both systems. The link capacity (maximum bit rate) is found to be limited by receiver noise, and to be the same, in both systems. Calculate the link length L, stating any approximations or assumptions made.

[6]

b) Estimate the bit rate in the links of part (a). Hence, show that for these links the shot noise will be significantly less than the receiver noise, assuming that the optical signal-to-noise ratio (SNR) equals 12 when considering receiver noise only.

[6]

c) If the optical receiver has a quantum efficiency of 1, prove that the shot noise and receiver noise will be equal if the received optical power equals the noise equivalent power squared divided by twice the photon energy.

[4]

d) Show that if an optical link is limited by shot noise, the number of electrons per bit in the photocurrent is simply equal to the square of the optical signal-to-noise ratio. State any assumptions or approximations made.

[4]

5. A silicon p-i-n photodiode has *p*, *i* and *n* doping levels respectively of N_A, $N_{D^-} = 10^{20}\ m^{-3}$ and N_{D^+}. The distances from the diode surface to the top and bottom of the intrinsic layer are $X_1 = 1.0$ μm and $X_2 = 5$ μm, respectively.

a) Find the optical attenuation coefficient that maximises the fraction of incident photons which are absorbed in the intrinsic layer of this device.

[4]

b) Find the value of N_{D^+} that allows the peak electric field magnitude to be only 15% greater than the magnitude at the bottom of the intrinsic layer, while only depleting 0.4 μm of the *n* layer. Then, find the value of N_A that allows these conditions to be achieved while only depleting 0.4 μm of the *p* layer.

[8]

c) Find the applied bias voltage *V* such that the conditions in (b) are achieved and the peak electric field magnitude is 2.5×10^5 V/m.

[4]

d) Discuss why it may be beneficial to minimise the distances to which the depletion region extends into the *n* and *p* layers.

[4]